

THESIS PROPOSAL

THESIS TITLE: BUILDING A STATISTICALLY VALIDATED DECISION SUPPORT SYSTEM FOR FOREIGN EXCHANGE TRADING WITH CALIBRATED UNCERTAINTY AND RISK CONTROL

Advisor: ASSOCIATE PROFESSOR NGUYEN DINH THUAN

Implementation period: 09/2026 to 12/2026 (16 weeks)

Students:

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Introduction:

Retail FX decision-support tools typically display a directional forecast and a confidence label whose empirical foundation is weaker than their presentation suggests: daily returns have resisted prediction since Meese and Rogoff [1], and technical-rule replications show in-sample profitability largely disappearing out of sample, absorbed by a momentum factor [2], [3]. The core difficulty is inference under a low signal-to-noise ratio with a large hypothesis space — evaluating thousands of candidate rules finds spurious relationships unless the hypothesis space is enumerated in advance and the family-wise error rate is controlled [4]–[6] — compounded by the fact that a negative finding is uninformative without its minimum detectable effect size (MDES). Predictability also differs by axis: the HAR model of realized volatility [7]–[9] gives well-established *magnitude* forecasts while *direction* remains contested [10], [11], so a system should state explicitly which axis it claims skill on.

Preliminary work has established feasibility through three pre-registered phases, each closed with a written decision record. A complete pipeline covers six pairs (EUR/USD, GBP/USD, USD/JPY, AUD/USD, USD/CAD, USD/CHF) over 2010–2025 (≈ 34.9 M one-minute bars); a production volatility layer, a three-class probability layer, and a risk layer run end to end through a web application updated daily. The extended HAR model attains test-set QLIKE 0.1585 vs. 0.2172 for the prior baseline, and 14 ML/DL alternatives plus 509 combinations fail to displace it from the Model Confidence Set [12]. On direction, 8,652 rule hypotheses across twelve discovery branches yield zero survivors after family-wise error control. A pre-registered causal-vs-validation feature-selection ablation is completed and frozen; its sealed-set evaluation was withheld because the pre-specified opening rule was not met.

This thesis is therefore as much an **engineering** problem as a statistical one: a decision support system whose every number is traceable to a pre-registered measurement, delivered as software that runs continuously rather than a script for a report. The data pipeline, computation layer, API, and web application (see *Proposed Method and System Architecture*) are a primary deliverable, developed across all sixteen weeks rather than assembled at the end.

Market Survey:

Existing retail tools were surveyed to state the contribution relative to what exists. **Signal marketplaces** (MQL5/MetaTrader, TradingView) publish track records from the vendor's own in-sample selection, with no multiple-testing correction. **Portfolio-analytics services** (Myfxbook, FX Blue) report realized statistics but no calibrated forward-looking probabilities. **Broker "confidence" widgets** attach a qualitative label with no scoring rule. None discloses the hypothesis space searched, the multiple-testing correction, a calibration guarantee, or a quantified probability of ruin — gaps identified in decision support systems generally [27] and specifically targeted here.

Research Objectives:

Design, implement, and validate a decision support system in which forecast skill is established per target axis under multiple-testing control, uncertainty is communicated via calibrated conformal prediction, and position sizing is constrained by an estimated probability of ruin. Three quantity groups are kept distinct: **predictive axes** (direction, magnitude, tail probability), **calibration diagnostics** (coverage, reliability), and the **decision layer** (VaR, ES, stop-out probability, sizing).

- **Axis-specific predictability.** Under one fixed protocol, determine on which axis out-of-sample skill exists, reported separately.
- **Model-complexity benchmarking.** Compare HAR-type, tree-based, recurrent, attention-based, and foundation-model alternatives under proper scoring rules and formal tests.
- **Incremental information value.** Test whether central-bank text, exogenous/cross-asset variables, and candlestick geometry [26] add value, and whether causal filtering beats ordinary selection.
- **Calibrated uncertainty.** Validate adaptive conformal prediction for target coverage under distribution shift.
- **Risk-controlled decisions.** Produce a per-session record: volatility, stop distance, stop-out probability, VaR/ES with backtests, ruin-constrained sizing.
- **Power-bounded negative findings.** Report the MDES for every negative result.
- **Engineering a production architecture.** Design and extend, across all sixteen weeks, a five-layer software architecture (pipeline, computation layer, versioned artefacts, API, web application) so every number stays traceable.

Research Scope:

Daily-horizon decision support for six major USD pairs. The 2010–2025 sample splits into training, validation (selection only), and a held-out development-test segment scored once. A **sealed evaluation set** — six cross pairs never used in development (EUR/GBP, EUR/JPY, GBP/JPY,

AUD/JPY, EUR/CHF, NZD/USD) plus 2026 data for all twelve pairs — opens exactly once after the configuration is frozen, with no retuning after.

The primary source is one-minute bar data, from which daily bars and five-minute realized measures are computed; supplementary sources include the central-bank/macro calendar, exogenous daily series, and tick-derived transaction costs. Horizons are one, five, and twenty sessions (one-session default). The study explicitly does **not** target a profitable trading strategy — execution, microstructure, and sub-daily timing are out of scope; causality claims are limited to Granger-sense temporal precedence.

Research Subjects:

The statistical validation of decision support outputs for FX under a low signal-to-noise ratio: how skill is attributed to an axis and an information layer; how multiple-testing control, power analysis, and sealed-set validation combine to produce replicable claims; how calibrated forecasts convert into risk limits with a measured ruin probability; and how this evidence should be presented so a user can separate what is measured from what is assumed.

Proposed Method and System Architecture:

Each item below is labelled **(implemented)** or **(proposed)**; nothing is a finished result until it appears with measured numbers in Expected Results.

Proposed Predictive and Risk Layers

- **Magnitude — extended HAR.** *(implemented)* Heterogeneous-autoregressive on realized variance [7], extended with measurement-error correction [8] and jump/semivariance decomposition [9], selected over 14 ML/DL challengers and 509 combinations under the Model Confidence Set [12].
- **Direction/probability — online expert combination.** *(implemented)* Hedge-style combination [13] of four sub-models, re-weighted every session; challenged by linear, tree, recurrent, attention, and foundation-model alternatives.
- **Causal discovery.** *(implemented — validation role)* Five triangulated methods decide which exogenous variable is trusted; using the effect to adjust the forecast directly was tested and found not yet effective (Deferred / Future Work).
- **Calibration — Mondrian-stratified ACI.** *(implemented)* [19], [20], 1.2-point maximum deviation vs. 2.4–3.2 for alternatives. *(proposed)* PID-controller online update [30] for below-target coverage during drawdown.
- **Risk/decision.** *(implemented)* Empirical-quantile VaR/ES with Kupiec [21], Christoffersen [22], dynamic-quantile [23] backtests and a consistent scoring function [24]; Kelly-derived sizing [25] capped by ruin probability. *(proposed)* Two pair-specific tail-risk recalibrations, each diagnosed but not yet frozen: covariate-shift-weighted conformal/EVT quantiles for USD/JPY's confirmed distribution drift, and change-point-isolated exclusion of USD/CHF's 2015 SNB shock window; plus a portfolio-correlation adjustment and a measured slippage coefficient.
- **Meta-labeling confidence layer.** *(proposed)* A secondary model that decides sizing/confidence rather than re-predicting direction [35]. After removing a circular-feature leak, preliminary skill

(Brier skill score) is positive in 3/6 pairs, rising to 5/6 when the feature set is augmented with an exogenous implied-volatility index — a candidate risk-layer addition, integration not yet finalized.

Causal Discovery Method

The exogenous layer needs a method conditioning on many candidates simultaneously, separating a direct link from a shared confound, under the same false-discovery control as the pattern families. The **official gate** is Granger-type F-testing under step-down Westfall–Young control [4]. Four independent methods triangulate the result — the answer to "which algorithm, and how do you know it is causality, not correlation":

Method	Question	Key result
PCMCI [18], [29]	Survives simultaneous conditioning?	VIX survives on every pair; gold-implied volatility (GVZCLS) loses significance once conditioned on VIX — the confound-vs-cause distinction pairwise Granger cannot make
Double ML [33]	Effect <i>size</i> , nonlinear nuisance model?	Positive, Holm-significant in 6/6 pairs — strongest result
Causal Forest [34]	Regime-dependent?	Heterogeneous but inconsistent across pairs — no rule generalises
CausalImpact [28]	Did <i>this</i> event move volatility?	6/20 ECB meetings significant after FDR (vs. ~5% under noise)

Four independently authored methods, different assumptions, converging on one variable is stronger evidence than any single p-value; none establishes structural causality — every claim remains temporal-precedence or potential-outcomes under its method's assumptions.

Two roles, one shipped. The **validation role is implemented and load-bearing** — it is the actual filter behind the frozen feature set. The **forecast-performance role** (adjusting the forecast directly) was tested and produced **no significant QLIKE improvement after Holm correction**, plausibly because HAR already absorbs the information indirectly; consistent with this thesis's negative-findings policy, it moves to Deferred / Future Work.

Proposed System Architecture

(Pipeline and API: implemented, operating daily; extensions below are proposed.) Five layers share one computation layer across two deployment targets, so the live site and thesis numbers share one code path:

1. **Data pipeline** ('collect/'). One-time historical bootstrap plus a daily job (four times/day via CI) building daily bars, realized measures, and the macro calendar.

2. **Data store** ('data/'). Frozen historical CSVs kept separate from a daily-overwritten live store,

so development and sealed segments cannot be silently mixed.

3. **Computation layer** (`src/`, `api/cache.py`). Series join, HAR forecast, calibration, and sizing run in one process, no serialisation boundary.

4. **API** (`api/`, **FastAPI**). One router per concern (meta, market, forecast, risk, admin) with Pydantic models, exposing endpoints extended through the Research Plan.

5. **Web application**. One HTML/JS template is the source of truth, compiled to three deployed variants.

The two deployment targets are a stated trade-off, not an inconsistency. The local/CI deployment runs the full FastAPI process and is the only place risk numbers are *computed*; Vercel's serverless size limit cannot host that stack, so a daily job pre-computes once-a-day numbers as static JSON while a lightweight function handles only intraday chart data — practice consistent with scaling ML-serving systems under heterogeneous constraints [31], [32].

Module contract, enforced in code, not an informal call:

From	To	Via	Format
collect/*.py	data/	direct write	fixed columns
data/	api/cache.py	fixed-path read	Date, open, ..., rv5
src/*.py	api/cache.py	in-process call	DataFrame, no serialisation
api/cache.py	api/routers/*.py	in-memory dict	pan, xs, sig, ...
api/routers/*.py	web client	HTTP JSON	response_model
daily job	static site data	HTTP call to API	identical JSON, never recomputed

The daily job never recomputes through a second code path — it calls the API and snapshots the response, the discipline [31] that keeps the static and live deployments from diverging, applied to every module added under the Research Plan.

Research Methodology:

- **Literature review.** HAR family and extensions; FX directional-predictability evidence; statistically sound pattern discovery and multiple-testing control; conformal prediction; VaR/ES backtesting and ruin-constrained sizing.
- **Data preparation and protocol fixing.** Construct bars and realized measures; verify contiguity and session conventions; fix the partition, metrics, multiple-testing procedure, and sealed-set policy in writing **before** any scoring; anti-leakage self-test (truncating information after t leaves the value at t unchanged).
- **Baseline and model comparison.** Extended HAR and online combination as baselines; compare against regularised linear, tree, recurrent/attention, and foundation models [11], [14] under an

identical information set; evaluate with QLIKE [15], CRPS, log/Brier [16]; test with Diebold–Mariano [17], superior predictive ability [5], and Model Confidence Sets [12].

- **Pattern, exogenous, and causal analysis.** Enumerate each discovery family's hypothesis space in advance, step-down max- T resampling [4]; condition on the strongest null. The exogenous/causal layer (*Causal Discovery Method*) is complete; remaining work is write-up (Week 9–10) — the forecast overlay stays deferred.
- **Calibration, risk, decision layer.** Mondrian ACI [19], [20]; VaR/ES backtests [21]–[24]; sizing as min(growth-optimal fraction [25], ruin-constrained cap), adjusted by regime, drawdown, correlation, and slippage.
- **Power analysis and sealed validation.** Inject synthetic effects through the actual funnel, report the MDES at 80% power; freeze the configuration, then open the sealed set exactly once.
- **System implementation.** Maintain the operating product path throughout, integrating each layer as a versioned artefact the week it completes; harden the system in the final weeks with full provenance; analyse failure cases and calibration degradation in the interface rather than omitting them.

Expected Results and Contributions:

- **A validated FX decision support system** issuing, per session and pair, a volatility forecast, calibrated three-class probability with a conformal set, event-risk assessment, and ruin-constrained sizing, every quantity traceable to a measurement.
- **An axis-resolved characterisation of predictability** — significant skill on magnitude, none on direction across twelve branches; remaining horizons and the sealed set (not yet opened) complete the picture.
- **A systematic model-complexity benchmark** for volatility forecasting across model families, with proper scoring rules and model confidence sets.
- **A five-method causal triangulation and ablation with an honest two-role split.** Causally-filtered variables degrade less out of sample than validation-selected ones (mechanism: distribution drift in the latter); the triangulation's validation role is load-bearing, its forecast-performance role tested and reported negative rather than overstated.
- **Power-bounded negative evidence** — every negative finding with its MDES, turning "no relationship found" into "no relationship stronger than X ".
- **Three applied risk-decision contributions:** a portfolio correlation factor for ruin-constrained sizing, a holding-horizon table against stop-out misreading, and a slippage coefficient measured from real executions.
- **A reproducible research record** — data dictionary, sealed-set protocol, full configuration registry, reproduction commands including negative results.
- **A documented, versioned software architecture** — the five-layer pipeline/API/web application with an explicit module contract, so the engineering contribution is itself reported and defensible.

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Research Plan and Timeline:

Research timeline: 16 weeks. Every week pairs a research task with an **Engineering** deliverable, so the API/web application is extended continuously rather than only at the end.

No.	Assignments	Timeline
1	– Consolidate the pipeline; fix the evaluation protocol; reproduce baselines. – Eng : verify /health//meta diagnostics; tag the architecture's starting version.	Week 1–2
2	– Complete pattern-discovery families at $h=5/20$; document the hypothesis space. – Eng : register results in the /models registry.	Week 3–4
3	– MDES analysis for every negative finding; validate the funnel with negative controls. – Eng : add MDES to /calibration.	Week 5–6
4	– Address tail-risk failures (two pairs, 99% VaR/ES); recalibrate $h=20$. – Eng : update /risk and /forecast_next fields.	Week 7–8
5	– Write up the completed exogenous/causal analysis, per axis. – Eng : expose causal-validation provenance as a read-only /forecast field.	Week 9–10
6	– End-to-end economic result after costs/slippage; evaluate the meta-labeling confidence layer as a candidate sizing adjustment. – Eng : extend /journal with rolling calibration.	Week 11
7	– Freeze the final configuration in writing; prepare the sealed-set record. – Eng : tag and freeze API schemas and the artefact version.	Week 12
8	– Open the sealed set once; score without retuning. – Eng : serve results through the same frozen endpoints.	Week 13
9	– Complete the system: computation layer, versioned artefacts, API, UI with full provenance; verify continuous operation.	Week 14
10	– Analyse behaviour by regime/pair/year; document failure cases. – Eng : add regime/pair/year breakdowns to the UI.	Week 15

11	– Complete the thesis report; revise per feedback; prepare the defence.	Week 16
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Deferred / Future Work (Not Scheduled):

- **Causal-conditioned forecast overlay.** The VIX effect is robust (6/6 pairs), but overlaying it on the HAR forecast gave no significant QLIKE gain — deferred pending a better functional form.
 - **Contemporaneous causal edges (PCMCI+, $\tau=0$).** Sign-ambiguous edges inconsistent with the frozen lagged result — needs a larger sample or a nonlinear test.
 - **Cross-pair volatility spillover (DYNOTEARS).** Real structure detected but does not improve test-set QLIKE as an added feature — not yet forecastable at a daily horizon.
 - **FX-specific implied volatility ('EVZCLS')** was discontinued in March 2025 and is unavailable.
- None of the above blocks the sealed-set evaluation or system hardening in Weeks 12–15.

Work Assignment:

(Percentage split confirmed against actual division of labour, not the undifferentiated 50/50 the advisor rejected. Both members remain jointly responsible for the whole system.)

Assignments	Member A (Nguyen Quoc Huy)	Member B (Huynh Tran Quoc Huy)
Review literature and define research scope	55%	45%
Construct the data pipeline and realized measures	40%	60%
Fix the evaluation protocol and sealed-set policy	50%	50%
Develop and benchmark the volatility forecasting layer	65%	35%
Develop the pattern-discovery and symbolic-representation branch	65%	35%
Implement multiple-testing control and stability screening	60%	40%
Conduct minimum detectable effect size and power analysis	60%	40%
Develop the exogenous, causal, and cross-asset information layers	65%	35%
Implement conformal prediction and calibration diagnostics	40%	60%
Develop the risk layer: VaR/ES backtesting and position sizing	55%	45%
Conduct the end-to-end economic evaluation	50%	50%
Execute the sealed-set validation	50%	50%
Design and implement the data pipeline scheduling and module contracts	35%	65%

Implement the API layer (FastAPI routers, response schemas)	30%	70%
Implement the web application and its three deployment variants	30%	70%
Analyse failure cases and document limitations	50%	50%
Write and revise the thesis report	50%	50%
Prepare for the final defence	50%	50%

Advisor's approval <i>(Signature and full name)</i> NGUYEN DINH THUAN	Ho Chi Minh City, ... / ... / Students <i>(Signatures and full names)</i> NGUYEN QUOC HUY HUYNH TRAN QUOC HUY
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